14. Proposed Application Enhancements

Based on the technical analysis of the existing application, several opportunities for improvement have been identified. These enhancements are intended to improve application reliability, data quality, error handling and maintainability without requiring major changes to the existing architecture.

14.1 HTTP Request Timeout Handling

Current observation:

The application uses HTTP GET requests to retrieve webpages. An explicit request timeout was not identified in the reviewed request calls.

Proposed improvement:

Introduce connection and read timeout settings for HTTP requests.

Implementation approach:

- Configure appropriate connection and read timeouts.
- Handle timeout exceptions separately.
- Record timeout-related failures in application logs.

Expected benefit:

Prevents requests from waiting indefinitely and improves reliability during network issues.

14.2 Retry Mechanism Improvement

Current observation:

The existing LoadPage() method supports a maximum of three attempts with random delays between retries.

Proposed improvement:

Improve the retry mechanism by introducing increasing wait intervals between failed attempts.

Implementation approach:

- Retain a configurable maximum retry count.
- Introduce exponential backoff.
- Handle temporary server errors and rate-limiting responses appropriately.
- Avoid repeatedly retrying permanent failures.

Expected benefit:

Improves recovery from temporary network and server-side failures.

14.3 Duplicate Research Paper Prevention

Current observation:

Duplicate paper links are removed during listing-page extraction. However, duplicate records may still occur across different collection dates or repeated executions.

Proposed improvement:

Introduce duplicate checking before adding records to the final dataset.

Implementation approach:

- Use the individual paper URL as a unique identifier.
- Maintain a collection of already processed URLs.
- Check existing CSV records before processing or appending new entries.

Expected benefit:

Prevents repeated records and improves the consistency of the collected dataset.

14.4 Extracted Data Validation

Current observation:

Extracted paper information is collected and added to the dataset before CSV generation.

Proposed improvement:

Introduce basic validation to identify incomplete or invalid records.

Implementation approach:

- Check whether mandatory fields such as paper title and individual paper URL are available.
- Validate publication and submission date formats.
- Identify missing optional fields such as abstracts and PDF links.
- Record incomplete entries for further review.

Expected benefit:

Improves data quality and reduces incomplete records in the final CSV.

14.5 Failed URL Tracking and Recovery

Current observation:

The application maintains separate lists for failed listing-page URLs and individual paper URLs.

Proposed improvement:

Introduce a structured failure report and an option to retry previously unsuccessful URLs.

Implementation approach:

- Record failed URLs, failure reasons and attempt counts.
- Maintain separate categories for listing-page and paper-page failures.
- Provide an option to process failed URLs again.
- Update the failure status after successful retrieval.

Expected benefit:

Makes failed collection operations easier to investigate and recover without repeating the complete process.

14.6 PDF Download and Availability Improvements

Current observation:

The Processor.py module contains a PDF download function, but its invocation in Main.py appeared to be commented out. A separate utility checks the availability of locally stored PDF documents.

Proposed improvement:

Improve PDF handling by introducing optional downloading and better download status tracking.

Implementation approach:

- Make PDF downloading configurable.
- Validate downloaded content before saving.
- Avoid downloading files that already exist.
- Record successful and unsuccessful downloads.
- Maintain download status for each paper.

Expected benefit:

Improves PDF management and makes missing or unsuccessful downloads easier to identify.

14.7 Additional Minor Improvements

The following improvements can also be considered:

Area| Proposed improvement| Expected benefit
Configuration| Store sensitive credentials using environment variables or an approved secrets-management solution.| Improved credential security
Logging| Include consistent timestamps, URLs and failure details.| Easier troubleshooting
CSV handling| Validate records and use safer file-writing practices.| Better data integrity
Code maintainability| Separate complex processing responsibilities into smaller methods where appropriate.| Easier maintenance
Configuration flexibility| Allow date ranges and request settings to be configured without modifying application logic.| Easier operation

15. Recommended Implementation Priority

The proposed enhancements can be prioritized as follows:

Priority| Enhancement| Reason
High| HTTP timeout handling| Prevents requests from waiting indefinitely
High| Retry mechanism improvement| Improves recovery from temporary failures
High| Duplicate prevention| Avoids repeated paper records
High| Data validation| Improves extracted data quality
Medium| Failed URL tracking and recovery| Simplifies failure investigation
Medium| PDF handling improvements| Improves PDF management
Medium| Configuration and logging improvements| Supports maintainability and security

The implementation order should be finalized based on team requirements and technical feasibility.

16. Conclusion

The Hugging Face Python application provides the basic functionality required for automated research-paper information collection.

The technical analysis identified its application architecture, execution workflow, data extraction process, supporting modules and existing limitations.

The proposed enhancements focus on improving webpage retrieval reliability, data validation, duplicate prevention, error recovery and PDF management.

These improvements can be introduced incrementally within the existing application structure without requiring a complete redesign.

The recommendations are intended for future development consideration and should be reviewed and prioritized based on team requirements.